

# Notes: Acemoglu, Ozdaglar, and Tahbaz-Salehi (AER, 2015)

## Systemic Risk and Stability in Financial Networks

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# 1 Big picture

- **Research question.** How does the architecture of an interbank financial network affect contagion and systemic risk? In particular, does making the network more complete and diversified make the system safer, or does it create a stronger channel for cascading failures?
- **Core answer.** The paper’s central result is a **phase transition**. When negative shocks are small, dense interconnections are stabilizing because losses are spread across many counterparties. When shocks are large, the same dense interconnections become destabilizing because they transmit distress to the whole system.
- **Economic mechanism.** There are two shock absorbers. First, non-distressed banks have excess liquidity, which is best used in a complete network. Second, senior outside creditors of a distressed bank can absorb losses, which is best used when the distressed bank is weakly connected to the rest of the system.
- **Main theoretical contribution.** The paper reconciles two apparently opposite views: the Allen–Gale view that diversification of interbank claims improves stability, and the contagion view that dense linkages amplify systemic risk. Both are right, but in different shock regimes.
- **Policy implication.** Counterparty exposure limits and network-design regulation cannot be evaluated without knowing the likely size and number of shocks. A rule that helps against small shocks may be harmful against large shocks.

# 2 Model and key objects

## 2.1 Environment

- The economy has  $n$  banks and three dates,  $t = 0, 1, 2$ .
- At  $t = 0$ , banks borrow from one another through unsecured debt contracts and invest in projects.
- Each project pays a random short-term return  $z_i$  at  $t = 1$  and, if held to maturity, a fixed non-pledgeable long-term return  $A$  at  $t = 2$ .
- The long-term return is not pledgeable, so all debts must be cleared at  $t = 1$ .
- Bank  $i$  also owes senior outside liabilities  $v$ , interpreted as obligations to depositors, secured creditors, taxes, wages, or other senior claimants.
- If a bank cannot meet its obligations at  $t = 1$ , it may liquidate its project. Liquidating  $\ell_i \leq A$  units yields only  $\zeta \ell_i$ , where  $\zeta \in [0, 1]$  captures liquidation efficiency.

## 2.2 Financial network

Let  $y_{ij}$  denote the face value of the debt owed by bank  $j$  to bank  $i$ . Thus,  $i$  is a creditor of  $j$ . The financial network is a weighted directed graph whose nodes are banks and whose directed weighted

edges represent interbank liabilities. Bank  $j$ 's total interbank liabilities are

$$y_j = \sum_{i \neq j} y_{ij},$$

while bank  $i$ 's total interbank claims are  $\sum_{j \neq i} y_{ij}$ .

- A network is **symmetric** if  $y_{ij} = y_{ji}$  for all pairs  $i, j$ .
- A network is **regular** if all banks have the same total interbank claims and the same total interbank liabilities, equal to  $y$ .
- The **ring network** is the sparsest regular benchmark: each bank owes all its interbank debt to one creditor.
- The **complete network** is the densest regular benchmark: each bank spreads its liabilities equally across all other banks, so  $y_{ij} = y/(n-1)$  for all  $i \neq j$ .

### 2.3 Convex combinations of networks

A network  $\{\tilde{y}_{ij}\}$  is a  $\gamma$ -convex combination of two networks  $\{y_{ij}\}$  and  $\{\hat{y}_{ij}\}$  if

$$\tilde{y}_{ij} = (1 - \gamma)y_{ij} + \gamma\hat{y}_{ij}, \quad \gamma \in [0, 1].$$

**Read:** This definition lets the paper move continuously from the ring network to the complete network. Increasing  $\gamma$  means liabilities become more diversified across counterparties.

## 3 Payment equilibrium: all equations and interpretation

### 3.1 Funds available before liquidation

Let  $x_{js}$  be the repayment made by debtor bank  $s$  to creditor bank  $j$ . Bank  $j$ 's cash flow before liquidation is

$$h_j = c_j + z_j + \sum_{s \neq j} x_{js},$$

where  $c_j$  is cash carried into  $t = 1$ .

### 3.2 Liquidation decision

If bank  $j$  does not have enough funds to cover senior liabilities  $v$  and interbank liabilities  $y_j$ , it liquidates assets according to

$$\ell_j = \left[ \min \left\{ \frac{1}{\zeta}(v + y_j - h_j), A \right\} \right]^+.$$

**Read:** The bank liquidates only if there is a shortfall. It liquidates just enough to fill the shortfall, unless the shortfall is too large, in which case it liquidates the whole project.

### 3.3 Repayments to junior interbank creditors

The payment by debtor bank  $j$  to creditor bank  $i$  is

$$x_{ij} = \frac{y_{ij}}{y_j} [\min\{y_j, h_j + \zeta \ell_j - v\}]^+.$$

**Read:** Senior outside creditors are paid first. Only the remaining resources are distributed to junior interbank creditors, and they are paid in proportion to the face values of their claims.

### 3.4 Payment equilibrium

A **payment equilibrium** is a collection of repayments and liquidation decisions,

$$(\{x_{ij}\}, \{\ell_i\}),$$

that simultaneously satisfies the liquidation and repayment equations for all banks.

- **Proposition 1.** For any financial network, cash holdings, and shock realization, a payment equilibrium exists and is generically unique.
- This extends the Eisenberg–Noe clearing-vector logic to a setting with different creditor seniorities and costly liquidation.

## 4 Social surplus and performance metrics

For most of the analysis, the paper assumes short-term project returns take two values:

$$z_i \in \{a, a - \epsilon\}, \quad a > v.$$

A bank hit by a negative shock receives  $a - \epsilon$  rather than  $a$ . The paper also restricts  $\epsilon$  so that the distressed bank has a positive short-run return, but, absent interbank repayments, would be unable to pay its senior outside creditors; with liquidation efficiency  $\zeta$ , this is captured by  $\epsilon \in (a - v + \zeta A, a)$ .

Conditional on  $p$  negative shocks, social surplus is

$$u = n(a + A) - p\epsilon - (1 - \zeta) \sum_{i=1}^n \ell_i.$$

When liquidation proceeds are trivial,  $\zeta = 0$ , surplus becomes

$$u = na - p\epsilon + (n - \# \text{ defaults})A.$$

**Read:** With trivial liquidation proceeds, maximizing surplus is equivalent to minimizing the number of defaults.

### 4.1 Stability and resilience

For two regular networks, conditional on  $p$  shocks:

- Network  $Y$  is **more stable** than  $\tilde{Y}$  if its expected surplus is higher.

- Network  $Y$  is **more resilient** than  $\tilde{Y}$  if its worst-case surplus is higher.

**Read:** Stability is an average-case concept; resilience is a worst-case concept.

## 5 Main theoretical results

### 5.1 Aggregate interbank liabilities

**Proposition 3.** If all pairwise interbank liabilities are scaled up by a factor  $\beta > 1$ , the financial system becomes less stable and less resilient.

**Why it matters:** More interbank debt increases each bank's exposure to counterparties. Even before studying network shape, the total scale of interbank liabilities matters for contagion.

### 5.2 Small-shock regime: diversification is stabilizing

Define the critical threshold

$$\epsilon^* = n(a - v).$$

This is the total excess liquidity of the financial network as a whole before the shock: absent shocks, each bank has  $a - v$  of liquidity after senior outside liabilities are met.

**Proposition 4.** Suppose  $\epsilon < \epsilon^*$  and interbank liabilities are sufficiently large,  $y > y^*$ . Then:

1. the ring network is the least stable and least resilient network;
2. the complete network is the most stable and most resilient network;
3. a  $\gamma$ -convex combination of the ring and complete networks becomes weakly more stable and resilient as  $\gamma$  increases.

**Read:** In the small-shock regime, spreading losses across more creditors lets the system use the excess liquidity of many non-distressed banks. The complete network diversifies the loss most effectively.

### 5.3 Majorization in the small-shock regime

The paper also studies transformations that make interbank liabilities more evenly distributed. The relevant transformation is an  $(\mathcal{D}, \mathcal{D}, P)$ -majorization: it mixes the liabilities of banks in default to the rest of the system using a doubly stochastic matrix, while keeping the resulting network regular.

**Proposition 5 and Corollary 1.** When  $\epsilon < \epsilon^*$  and  $y > y^*$ , if the distressed bank can meet its senior outside liabilities, this majorization does not increase the number of defaults. If a network has no contagion, any  $\gamma$ -convex combination with the complete network also has no contagion.

**Why it matters:** This formalizes the Allen–Gale intuition: diversification of interbank exposure is protective when shocks are not too large.

### 5.4 Large-shock regime: diversification becomes contagious

A regular financial network is  $\delta$ -connected if there exists a subset of banks  $\mathcal{S} \subset \mathcal{N}$  such that

$$\max\{y_{ij}, y_{ji}\} \leq \delta y \quad \text{for all } i \in \mathcal{S}, j \notin \mathcal{S}.$$

For small  $\delta$ , the network contains a weakly connected component.

**Proposition 6.** Suppose  $\epsilon > \epsilon^*$  and  $y > y^*$ . Then:

1. the complete and ring networks are the least stable and least resilient networks;
2. for small enough  $\delta$ , any  $\delta$ -connected network is strictly more stable and resilient than the ring and complete networks.

**Read:** Once the shock is too large for the system's excess liquidity to absorb, the complete network transmits losses to everyone. Weakly connected networks can contain contagion because the distressed bank's senior creditors absorb more of the loss.

**Takeaway:** Dense financial networks are **robust-yet-fragile**: robust to small shocks, fragile to large shocks.

## 5.5 Large-shock majorization

**Proposition 7 and Corollary 3.** In the large-shock regime, a majorization that further diversifies the liabilities of defaulting banks does not reduce the number of defaults. Relatedly, for sufficiently high bottleneck networks, taking a  $\gamma$ -convex combination with the complete network is no more stable or resilient.

**Why it matters:** The same type of diversification that helps under small shocks can fail under large shocks. Diversification spreads losses that would otherwise be borne by the distressed bank's senior creditors.

## 6 Harmonic distance and systemic importance

### 6.1 Harmonic distance

For bank  $i$  and distressed bank  $j$ , the paper defines harmonic distance by

$$m_{ij} = 1 + \sum_{k \neq j} \left( \frac{y_{ik}}{y} \right) m_{kj}, \quad m_{ii} = 0.$$

**Read:** This distance is not simply shortest-path distance. It accounts for all liability chains that connect two banks and weights them by the intensity of interbank liabilities.

**Proposition 8.** Suppose bank  $j$  is hit by a large shock and defaults on senior liabilities. Then there exists  $m^*$  such that:

1. if  $m_{ij} < m^*$ , bank  $i$  defaults;
2. if all banks default, then  $m_{ij} < m^*$  for all  $i$ .

**Why it matters:** Banks that are closer to the distressed bank in harmonic distance are more vulnerable to default. A bank is systemically important if it is harmonically close to many others.

### 6.2 Bottleneck parameter

The bottleneck parameter is

$$\phi = \min_{\emptyset \neq S \subsetneq \mathcal{N}} \frac{\sum_{i \in S} \sum_{j \notin S} (y_{ij}/y)}{|S| |\mathcal{S}^c|}.$$

**Read:** The parameter measures how difficult it is to cut the network into two weakly connected parts. Complete networks have high bottleneck parameters; disconnected or weakly connected networks have low bottleneck parameters.

**Lemma 1.** For any symmetric financial network,

$$\frac{1}{2n\phi} \leq \max_{i \neq j} m_{ij} \leq \frac{16}{n\phi^2}.$$

**Corollary 2.** In the large-shock regime, high bottleneck networks are more likely to suffer universal default, while low bottleneck networks can preserve at least one surviving bank.

## 7 Extensions

### 7.1 Multiple shocks

If  $p$  banks are hit by negative shocks, the critical threshold becomes

$$\epsilon_p^* = \frac{n(a - v)}{p}.$$

**Proposition 9.** The same phase-transition logic applies with multiple shocks, but with an additional intermediate-liability case:

1. if  $\epsilon < \epsilon_p^*$  and  $y$  is sufficiently large, the complete network is the most stable and resilient, while the ring network is the least resilient;
2. if  $\epsilon > \epsilon_p^*$  and  $y$  is sufficiently large, the complete and ring networks are the least stable and resilient, and a sufficiently weakly connected network can be strictly safer when  $p < n - 1$ ;
3. if  $\epsilon > \epsilon_p^*$  but  $y$  lies in an intermediate range, the complete network is the least stable and resilient, and the ring can be strictly more stable than the complete network.

**Why it matters:** More shocks reduce the size of the shock each bank can absorb. The number of shocks plays a role similar to shock magnitude, and with multiple shocks the senior creditors in a ring can sometimes absorb losses more effectively than in the complete network.

### 7.2 Positive liquidation proceeds

When liquidation produces nontrivial proceeds,  $\zeta > 0$ , the relevant thresholds become

$$\epsilon_*(\zeta) = n(a - v) + \zeta A, \quad \epsilon^*(\zeta) = n(a - v) + n\zeta A.$$

**Proposition 10.** With partial liquidation and  $\zeta > 0$ , the phase transition becomes smoother:

1. if  $\epsilon < \epsilon_*(\zeta)$ , the complete network is the most stable and resilient and the ring is the least stable and resilient;
2. if  $\epsilon > \epsilon^*(\zeta)$ , the complete and ring networks are the least stable and resilient, while sufficiently weakly connected networks are strictly safer;



3. if  $\epsilon_*(\zeta) < \epsilon < \epsilon^*(\zeta)$ , the complete network is strictly more stable and resilient than the ring, but once  $\epsilon > \epsilon_*(\zeta) + \zeta A$ , some sufficiently weakly connected network can be strictly more stable and resilient than the complete network.

**Proposition 11.** If partial liquidation is not feasible, the sharp phase transition returns even when  $\zeta > 0$ : below  $\epsilon_*(\zeta)$  the complete network is safest, while above  $\epsilon_*(\zeta)$  the complete and ring networks are least stable and resilient.

### 7.3 Size heterogeneity

If all assets and liabilities of bank  $i$  are scaled by  $\theta_i > 0$ , systemic fragility depends on a size-adjusted harmonic distance:

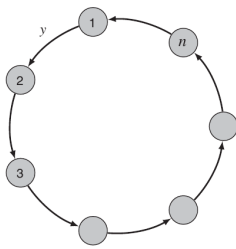
$$\hat{m}_{ij} = \theta_i + \sum_{k \neq j} \left( \frac{y_{ik}}{y_i} \right) \hat{m}_{kj}.$$

**Proposition 12.** If bank  $j$  receives a sufficiently large shock, then  $j$  defaults on its senior liabilities, and all other banks default if and only if  $\hat{m}_{ij} < \theta_i m^*$  for all  $i$ . Increasing the relative size of an intermediary bank  $k \neq i, j$  makes bank  $i$  more robust to a shock to  $j$ , but increasing bank  $i$ 's own relative size can make it more vulnerable because the default threshold  $\theta_i m^*$  rises faster than the size-adjusted distance  $\hat{m}_{ij}$ .

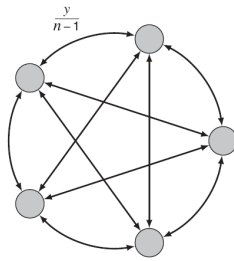
## 8 Figures from the paper: original titles and numbers

The paper contains three figures and no tables. The captions below keep the original figure numbers and titles from the paper. Adjacent figures are placed side-by-side where possible.

Panel A. The ring financial network

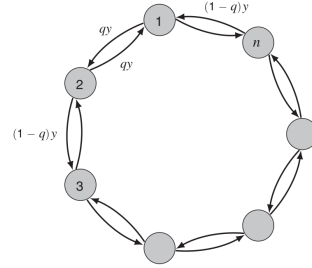


Panel B. The complete financial network



**Figure 1. The Ring and the Complete Financial Networks**

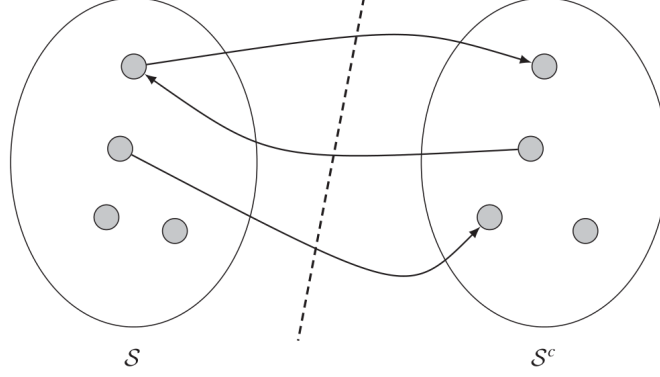
**Read:** Panel A is the sparsest regular benchmark; Panel B is the densest regular benchmark. In the complete network, liabilities are spread equally across counterparties.



**Figure**

**2. The Liabilities of Each Bank to Its Two Counterparties Become More Equalized as  $q$  Approaches 1/2**

**Read:** This example shows that not every equalization of liabilities lowers fragility. Redistribution can move resources away from the bank that needs them to survive.



**Figure 3. A Partition of the Financial Network into Subsets  $\mathcal{S}$  and  $\mathcal{S}^c$**

*Note:*  $\sum_{i \in \mathcal{S}, j \notin \mathcal{S}} y_{ij}$  is equal to the sum of interbank liabilities over the dashed line and measures the aggregate liabilities of banks in  $\mathcal{S}^c$  to those in  $\mathcal{S}$  and vice versa.

**Read:** The dashed line illustrates the bottleneck parameter. A high-bottleneck network is hard to split into weakly connected components; a low-bottleneck network has a natural point of separation.

## 9 Short summary

- The paper studies systemic risk in a network of banks connected by unsecured interbank debt.
- Payment equilibrium exists and is generically unique, even with creditor seniority and costly liquidation.
- Increasing the scale of interbank liabilities makes the system more fragile.
- With small shocks, dense networks are stabilizing because they spread losses and use excess liquidity efficiently.
- With large shocks, dense networks are destabilizing because they transmit losses throughout the system.
- Weakly connected networks can be safer under large shocks because senior outside creditors absorb more losses.
- Systemic importance is not standard eigenvector or Bonacich centrality; it is captured by harmonic distance in the financial network.

## 10 Conclusion

**Core conclusion.** The relationship between interbank connectivity and systemic risk is non-monotone. More connectivity is good in one regime and bad in another.

**Economic intuition.** Complete networks diversify small losses, but they also make every bank exposed to large losses. Weak links are inefficient for sharing small shocks but useful for containing large shocks.

**Policy takeaway.** Network regulation should not simply aim for either more diversification or fewer interconnections. The optimal policy depends on the expected size and multiplicity of shocks, the seniority structure of liabilities, and the harmonic-distance position of systemically important banks.